

Filters Introduction

Learning Objectives

- Understand what filters do to images
- Apply blur, sharpen, and noise filters
- Use the preview feature before applying filters

Engage (5-7 minutes)

When you look through a frosted glass window, the world outside looks blurry. Filters in GIMP are like different types of glasses—they transform how an image looks. Some make it sharper, some softer, some add artistic effects. Today we try on different digital glasses.

Explore (10-12 minutes)

Open an image in GIMP. Go to Filters > Blur and try Gaussian Blur—adjust the blur radius. Then try Filters > Enhance > Sharpen. Compare before and after using the preview checkbox. Try adding noise with Filters > Noise and observe the effect.

Explain (10-12 minutes)

Filters apply mathematical transformations to image pixels. Blur filters soften images by averaging nearby pixels. Sharpen filters increase contrast between adjacent pixels to make details stand out. Noise filters add random grain for texture or artistic effect. Always use the preview option before applying a filter to see the result, and use Undo (Ctrl+Z) if you do not like it.

Elaborate (8-10 minutes)

Think of looking through different types of glass—a clear window (no filter), frosted glass (blur), magnifying glass (sharpen), and textured glass (noise). Each changes how you see the world. Filters do the same to images. Why would you use blur on a background but sharpen on a portrait?

Evaluate (5-8 minutes)

Exit ticket: Name three types of filters and describe the effect of each. Then apply a blur filter to an image and undo it. If you finish early, try the Pixelize filter for a mosaic effect.